

Use Case Template

ChargEV – Smart EV Charging Station Platform

1. Use Case Title	Plan Multi-Stop Trip
2. Abbreviated Title	Plan Trip
3. Use Case Id	UC-07
4. Actors	EV Driver (Primary), Location Services API (Supporting)
5. Description	The EV Driver plans a long-distance trip that automatically chains together charging stops based on the vehicle's range.
5.1. Pre Conditions	1. Driver has specified an origin, a destination and the vehicle's range.
5.2. Task Sequence	1. Driver enters the start point, end point and vehicle range. 2. System computes the base route via the Directions API. 3. System identifies charging stations along the route at range-appropriate intervals. 4. System suggests an optimal set of charging stops for the trip.
5.3. Post Conditions	1. A multi-stop itinerary with suggested charging stops is presented to the driver.
6. Modification History	Date: September 18, 2026
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